Sprint 077: Thermodynamic State Vector Discrepancy Aggregator in Biosphere Simulation

Lead Scientific Communications & Academic Outreach Agent
Web of Life Project

March 2025

Abstract

Simulating complex biospheres requires rigorous enforcement of thermodynamic laws to maintain long-term stability and physical realism. Sprint 077 introduces the Thermodynamic State Vector Discrepancy Aggregator within `src/thermodynamics/state_validator.ts`. This component formalizes the evaluation pipeline by mapping state vectors across biogeochemical evaluations and computing maximum discrepancy accumulation vectors. By monitoring conservation bounds across Carbon, Nitrogen, Phosphorus, Water, and Enthalpy stocks, the module flags potential First Law violations and Second Law entropy generation anomalies. This paper details the architectural placement, mathematical formulations, interface contracts, and verification strategies deployed in the Web of Life framework.

Official Repository: `https://github.com/pascalranoroarijaona/WebOfLife`

1 Introduction and Thermodynamic Foundations

The Web of Life simulation models dynamic ecosystems where matter and energy flow continuously between trophic levels and abiotic pools. To prevent unphysical matter creation or energy leakage, the simulation architecture enforces foundational physical laws:

- **First Law of Thermodynamics:** Conservation of total mass and energy across stocks:
$$\sum \Delta\text{Stock}_{\text{in}} = \sum \Delta\text{Stock}_{\text{out}} + \Delta\text{Dissipation}$$
- **Second Law of Thermodynamics:** Bounding system degradation and unmodeled dissipation via entropy generation ($\Delta S_{\text{universe}} \geq 0$).

2 Architectural Placement and Class Hierarchy

The discrepancy aggregator extends the thermodynamic validation monad stack:

[BaseMonadProcess] $\longrightarrow$ [ThermodynamicMonadProcess] $\longrightarrow$ [StateValidator] $\longrightarrow$ [StateVectorDiscrepancyAg

Each conserved state vector $\mathbf{X}$ tracks five fundamental physical and chemical stocks:

1. **Carbon** (C) [kg]
2. **Nitrogen** (N) [kg]
3. **Phosphorus** (P) [kg]
4. **Water** (W) [kg]
5. **Enthalpy** (H) [J]

3 Mathematical Formulations

For any given thermodynamic evaluation i, the discrepancy δ_i is defined as the ℓ_2-norm of the difference between the expected state vector $\mathbf{X}_{\text{expected},i}$ and the actual measured state vector $\mathbf{X}_{\text{actual},i}$:

$$\delta_i = ||\mathbf{X}_{\text{expected},i} - \mathbf{X}_{\text{actual},i}||_2 = \sqrt{\sum_{k \in \{C,N,P,W,H\}} (X_{\text{expected},i,k} - X_{\text{actual},i,k})^2}$$

The aggregator computes the maximum discrepancy across an array of n evaluation results to bound unmodeled dissipation or entropy generation anomalies:

$$\Delta_{\max} = \max_{1 \le i \le n} (\delta_i)$$

4 Interface Contracts and Implementation

The TypeScript implementation within `src/thermodynamics/state_validator.ts` adheres to the following interface contracts:

```
export interface IStateEvaluationResult {
  timestamp: number;
  expectedVector: IStateVector;
  actualVector: IStateVector;
  discrepancy: number;
}

export interface IStateVectorAggregator {
  mapEvaluations(results: IStateEvaluationResult[]): number[];
  accumulateMaxDiscrepancy(results: IStateEvaluationResult[]): number;
}
```

5 Verification and Testing Strategy

Verification protocols for Sprint 077 include:

1. **Mapping Correctness:** Validating that `mapEvaluations` accurately projects evaluation results into scalar discrepancy arrays.
2. **Maximum Accumulation:** Verifying that `accumulateMaxDiscrepancy` extracts the correct supremum across stochastic simulation cycles.
3. **Conservation Bounds:** Confirming that zero-discrepancy states yield 0.0, while anomalies scale proportionally with injected mass-energy discrepancies.

6 Conclusion

Sprint 077 establishes a robust, mathematically grounded mechanism for monitoring thermodynamic integrity within the Web of Life biosphere simulation. By enforcing strict tracking of state vector discrepancies, the simulation maintains high fidelity to physical conservation laws.